

Simple Oxygen Concentrator You Can Assemble At Home

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In the present world, highly affected by pandemic, there is a huge spike in Covid-19 infection cases in many places. With limited resources the fight against Covid-19 virus is an unexpected challenge for the whole world. As we all saw, arranging oxygen cylinders for loved ones became a difficult task a few months ago in India. The patients and their close relatives suffered badly, instead of providing moral support to the patients, when they could not arrange oxygen cylinders.

To get rid of this situation, this DIY project tries to help through details about assembling an oxygen concentrator at home. With the help of easily available parts, one can make an oxygen concentrator with enough capacity to support a Covid-19 or any other patient in need of oxygen.

Circuit and working

This oxygen concentrator can be built using an Arduino Uno, 12V DC power supply, control relay, solenoid valves, air compressor, air filtration unit, flow control valve, membrane housing as canisters, PU fittings, and zeolite sieves.

Electronic control circuit is the main part of the oxygen concentrator. Oxygen is supplied (to the patient) through the solenoid valve outlets, which are controlled by the circuit presented here.

The circuit diagram of oxygen concentrator controller is shown in Fig. 1. It is built around 15V AC step-down transformer X1, bridge rectifier BR1, 12V voltage regulator (7812) IC1, Arduino Uno Board1, solenoid valves V1 through V6, 5mm LED (LED1), 2N2222 transistors T1 through T6, and 12V relays RL1 through RL6.

Arduino Uno. Arduino Uno board is connected to control relays. It is preprogrammed with delays

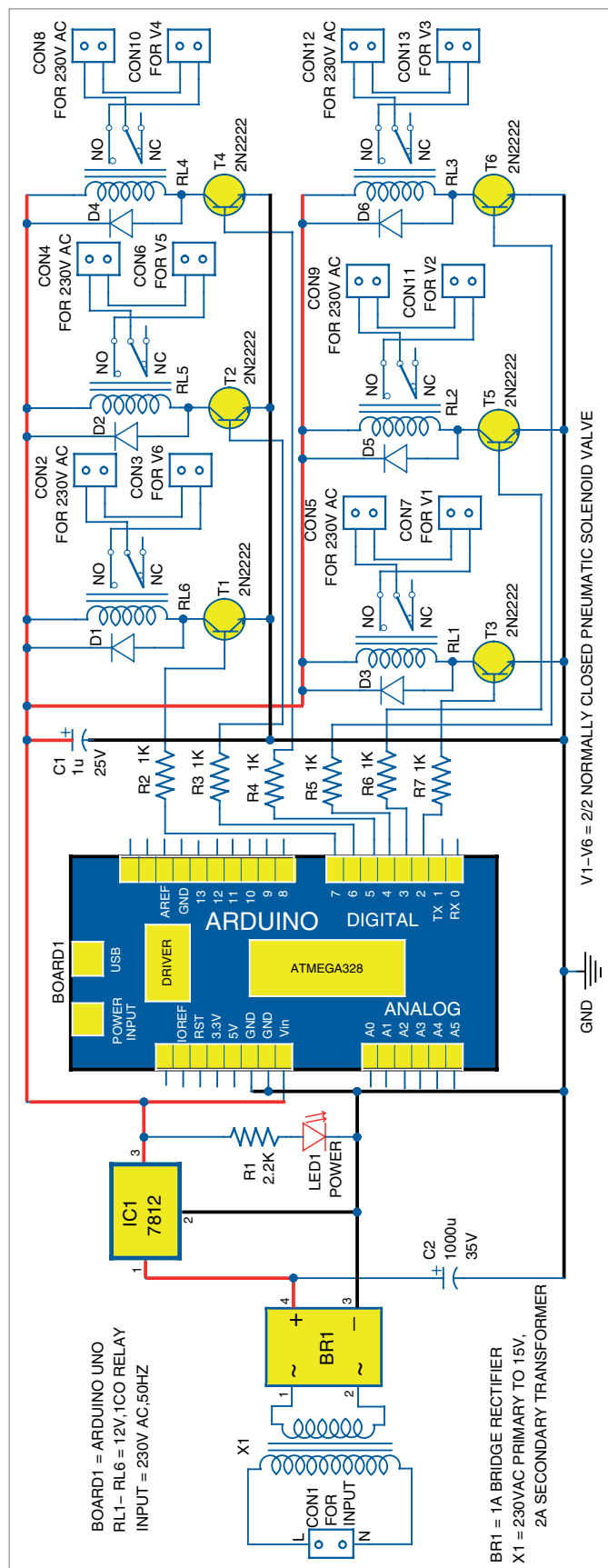


Fig. 1: Circuit diagram of oxygen concentrator controller